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			Effective Date	January 26, 2023
			Page	Page 1 of 8

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	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 2 of 8

(9A) Abstract

This study evaluates the proximate, mineral, and energy contents of locally available crop by-products, leaf meals, legumes, grains, sprouts, fish silage, and shrimp meal. The analysis reveals significant variations in nutrient composition, with rice bran and corn bran showing differences in dry matter, ash, protein, and fiber content. Leaf meals, notably moringa and madre de agua, offer high protein and mineral content, making them valuable for livestock nutrition. Legumes and sprouts, especially when sprouted, enhance protein levels in feed. Fish silage and shrimp meal provide balanced nutrient profiles, with shrimp meal showing higher overall nutrient content. These findings underscore the importance of tailored feed formulations and the potential of incorporating diverse feed ingredients to meet the specific nutritional needs of livestock.

(9B) Keywords: Locally available feedstuffs, leaf meals, crop by-products, sprouts, fish silage, proximate analysis

(10) Introduction

The increasing demand for sustainable and cost-effective animal feed has driven researchers and industry professionals to explore alternative nutrient sources. Traditional feed ingredients such as soybeans, fish meal, corn, and other common commercially available feedstuffs are becoming less sustainable due to their high cost and environmental impacts. The cultivation of these ingredients involves significant land use, water consumption, and greenhouse gas emissions, while fish meal production raises concerns about overfishing and marine ecosystem degradation (Van Duinkerken & Bannink, 2018). This growing challenge underscores the need to explore and assess alternative feed sources that can offer comparable or superior nutritional value while mitigating environmental and economic pressure.

Poultry and swine are major sources of animal protein and are extensively reared across diverse environments. Poultry, including chickens and ducks, and swine are valued for their rapid growth rates and efficient feed conversion (Kong & Yin, 2016). Their dietary needs are well-documented, making them ideal for evaluating new feed ingredients. The National Research Council (NRC) provides specific nutrient requirements for these animals, detailing the essential amino acids, protein, energy, vitamins, and minerals needed for optimal growth and health (NRC, 1994; NRC, 2012). Understanding how alternative feedstuffs meet these requirements is crucial for ensuring their effectiveness. In tropical countries, like Philippines, marine by-products, leaf meals, grains, and agricultural crop by-products are often abundant but underutilized. Siddiqui and Khan (2019) emphasize that alternative feed ingredients, including agro-industrial by-products and underutilized plant species, can offer significant benefits if their nutrient profiles are properly assessed.

Evaluating the nutrient composition of these locally available alternative feedstuffs relative to NRC's established requirements is essential for determining their effectiveness in meeting the nutritional needs of poultry and swine. This analysis is crucial for validating the viability of these alternative feeds as replacements for conventional ingredients, which can optimize their use and reduce reliance on imported or expensive feed resources. Integrating locally available agricultural by-products and other alternative feeds can enhance the sustainability of animal production systems by minimizing environmental impacts and waste and improving the efficiency of livestock operations (Khan & Zhang, 2020). Furthermore, this approach supports local agricultural economies, addresses the growing demand for animal protein, and contributes to food security and economic stability in tropical and developing countries where traditional feed resources may be less accessible or more costly. Hence, this study was conducted to evaluate and compare the proximate nutrient composition of various alternative feedstuffs and their suitability for use in poultry and swine diets.

(11) Materials and Methods or Methodology

Preparation of the feedstuffs

Crop and crop by-products were gathered in the nearby rice mill. Leaf meals were gathered from the animal farm of Capiz State University in Pontevedra. Corn were purchased from the nearby farmer. Sprouts were

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 3 of 8

prepared according to the method of Gacutan et al. (2022). Fish silage was prepared following the method of Escala et al. (2021).

Samples were partially dried and sent to Lipa Quality Control laboratory for analysis of the nutrient and energy contents.

(12) Results and Discussions

PROXIMATE, MINERAL COMPOSITION, AND ENERGY CONTENTS OF LOCALLY AVAILABLE CROP AND CROP BY-PRODUCTS

Proximate and mineral composition of common crop and crop by-products are presented in Table 1. The dry matter (DM) content for rice bran D2 (89.50%) aligns well with the NRC standard range of 89-91%. In contrast, rice bran D1 (85.54%) is below this range, indicating a higher moisture content. Corn bran (88.01%) is slightly below the NRC standard of 88-90%, suggesting a minor deviation in dry matter content. The ash content for rice bran D2 (10.64%) exceeds the NRC standard of 9-10%, whereas rice bran D1 (8.19%) is slightly below this range. Corn bran (2.89%) and cassava tuber (1.70%) have significantly lower ash content compared to the NRC standards of 3-5% and 1.5-2%, respectively, highlighting variability in mineral content, which is crucial for balancing dietary minerals.

Rice bran D1 (13.07%) meets the NRC standard of 12-14% for crude protein (CP), while rice bran D2 (10.03%) and corn bran (10.15%) are slightly below their respective standards of 10-12%. Cassava tuber (2.58%) is significantly lower than the NRC standard, indicating the need for protein supplementation when using these by-products in animal feed formulations. The crude fiber (CF) content in rice bran D2 (19.13%) is exceptionally high compared to the NRC standard of 7-10%, whereas rice bran D1 (6.23%) is slightly below this range. Corn bran (2.53%) and cassava tuber (1.98%) are considerably lower than the NRC standard of 8-12%, suggesting variability in fiber content which can affect the digestibility and utilization of these feeds.

The ether extract (EE) content in rice bran D1 (8.57%) is below the NRC standard of 12-14%, and rice bran D2 (0.66%) is significantly lower. Corn bran (1.59%) and cassava tuber (0%) also fall below their respective standards of 3-4%, indicating lower fat content, which suggests the need for additional energy sources when using these by-products. Both rice bran D1 and D2 (0.31%) exceed the NRC standard of 0.06-0.10% for calcium (Ca), while corn bran (0.21%) and cassava tuber (0.31%) exceed their respective standards. This highlights the importance of careful mineral balancing to prevent deficiencies or excesses.

For phosphorus (P), rice bran D1 (2.01%) exceeds the NRC standard of 1.2-1.8%, while rice bran D2 (0.97%) is below it. Corn bran (0.59%) and cassava tuber (0.04%) are also below their standards, indicating the need for supplementation to ensure adequate phosphorus levels in animal diets. Rice bran D1 (0.96%) exceeds the NRC standard of 0.6-0.7% for magnesium (Mg), while rice bran D2 (0.46%) is below it. Corn bran (0.19%) and cassava tuber (0.10%) fall below the standard range of 0.2-0.3%, underscoring the importance of mineral supplementation.

Rice bran D1 (1.41%) is within the NRC standard of 1.5-1.8% for potassium (K), whereas rice bran D2 (0.81%) is below it. Corn bran (0.64%) and cassava tuber (0.57%) are below the standard, suggesting the need for monitoring and adjusting potassium levels in diets. The gross energy (GE) values range from 3187.80 to 3522.50 kcal/kg for rice bran and 3200-3500 kcal/kg for corn bran, generally falling within the expected range for these by-products. The variability in GE values underscores the importance of selecting feed ingredients based on the specific energy needs of livestock.

For poultry, rice bran D1 (2994.13 kcal/kg) falls within the NRC standard of 2900-3100 kcal/kg, while rice bran D2 (2550.24 kcal/kg) is lower. For swine, rice bran D1 (2465.75 kcal/kg) is slightly below the NRC standard of

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 4 of 8

2800-3000 kcal/kg. Corn bran and cassava tuber show varied metabolizable energy (ME) values, highlighting the importance of selecting appropriate feed ingredients based on specific energy needs. The observed values and comparisons align with NRC standards, underscoring the variability and nutritional potential of local feed ingredients. Significant p-values ($p < 0.001$) across multiple nutrients emphasize the differences in composition, crucial for formulating balanced diets for livestock.

COMPARISON OF THE NUTRIENT COMPOSITION AND ENERGY CONTENTS OF LOCALLY AVAILABLE LEAF MEALS

Table 2 outlines the nutrient composition of leaf meals such as moringa, cassava leaves, and madre de agua. Results reveal significant differences ($p < 0.001$) in all parameters except DM. Madre de agua has the highest ash and calcium content, suggesting its potential as a mineral-rich feed supplement. Moringa leaves exhibit the highest crude protein content, making them a valuable protein source. The energy content shows that moringa and cassava leaves are comparable, both providing substantial energy levels for livestock diets. These findings support the inclusion of leaf meals in livestock diets to enhance nutrient diversity and meet specific dietary needs. For example, moringa's high protein content can be particularly beneficial in poultry diets where high protein is essential for growth and egg production.

PROXIMATE ANALYSIS, MINERAL COMPOSITION, AND ENERGY CONTENTS OF LOCALLY AVAILABLE LEGUMES, GRAINS, AND SPROUTS

Comparison of the proximate, mineral, and energy contents of locally available grains and sprouts presented in Table 3 showed significant differences. Soybean meal emerges as a superior protein source with significant crude protein content, while mungbean sprouts demonstrate a progressive increase in protein content over time. Corn and its sprouts show relatively lower protein but higher ether extract and gross energy values, indicating their primary role as energy sources in feed formulations.

These results highlight the potential of sprouts as enhanced nutritional feed ingredients. The increase in protein content over sprouting days suggests that sprouting grains could be an effective method to boost protein levels in animal feed, aligning with studies on improved nutrient bioavailability through sprouting.

COMPARISON OF THE PROXIMATE, MINERAL, AND ENERGY CONTENTS OF FISH SILAGE AND SHRIMP MEAL

Result of the comparative analysis of fish silage and shrimp meal presented in Table 4 showed significant differences ($p < 0.001$) in all parameters except in CP. Local shrimp meal shows has higher nutrient content compared to fish silage except in crude fat (EE) and potassium (K). CP of local shrimp meal and fish silage is comparable although values are lower to that of the NRC. Nevertheless, Fish silage offers a balanced nutrient profile with moderate levels of protein and minerals. Marine by-products are crucial in animal nutrition, particularly in aquaculture and poultry feed, due to their high-quality protein and essential fatty acids. The high crude protein in fish meal supports its use in high-performance diets where protein needs are substantial.

(13) Conclusions and Recommendations

CONCLUSION

This study provides a comprehensive analysis of the proximate, mineral, and energy contents of locally available crop by-products, leaf meals, legumes, grains, sprouts, fish silage, and shrimp meal. The findings reveal notable variations in nutrient composition and energy content among different feed ingredients. Rice bran and corn bran exhibit variability in dry matter, ash, protein, and fiber content, which highlights the need for careful formulation

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 5 of 8

to meet specific dietary requirements for livestock. Leaf meals, such as moringa and madre de agua, show significant potential for enhancing nutrient density, with moringa leaves being particularly valuable for their high protein content and madre de agua for its mineral richness. Legumes and sprouts offer promising alternatives for improving protein and energy levels in animal feeds, with sprouting shown to increase protein content.

Fish silage and shrimp meal present balanced nutrient profiles, with shrimp meal having higher nutrient content overall, though both are comparable in crude protein. These marine by-products are critical in high-performance feeds due to their high-quality protein and essential fatty acids.

RECOMMENDATIONS

1. Due to the variability in nutrient content, it is crucial to adjust feed formulations to ensure balanced diets for livestock, particularly when using rice bran, corn bran, and cassava tuber. Supplementing with additional protein or energy sources may be necessary.
2. Incorporate leaf meals like moringa and madre de agua into livestock diets to leverage their high protein and mineral contents. This can enhance the nutritional profile of feeds, especially for poultry and ruminants.
3. Utilize sprouted grains as they show increased protein content over time, making them a valuable addition to animal feed for improving protein levels.
4. Continue using fish silage and shrimp meal in high-performance diets due to their balanced nutrient profiles. Ensure that the protein and mineral contents of these by-products align with specific nutritional needs.
5. Regularly test and analyze locally available feed ingredients to account for variability and ensure that nutrient requirements are consistently met.

(14) Output

Information of the nutrient composition of different alternative nutrient sources for poultry and swine

(15) Technology Transfer Activity Conducted

Fact sheets, brochures, and technical bulletins that summarizes the detailed information on the nutrient composition and usage guidelines of this alternative feed ingredients

(16) Economic Analysis

The direct costs associated with implementing the project includes:

1. Labor for gathering the samples and preparation for analysis- 3 months @11,065=33, 195.00
2. Laboratory analysis

(17) Acknowledgments

This study was made possible through the support and funding provided by Capiz State University's Research, Development, and Extension Office to the Livestock Research and Development Center. Their generous financial assistance and unwavering support have been instrumental in the successful completion of this research.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 6 of 8

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 7 of 8

Crop and Crop By-products

Table 1. Proximate, mineral composition, and energy contents of locally available crop and crop by-products

Items (% DM)	Feedstuffs						SEM	P-value
	¹ Rice Bran	Rice Bran D ₁	Rice Bran D ₂	*Corn Bran	Corn Bran	Cassava Tuber		
DM	89-91	85.54 ^d	89.50 ^a	88-90	88.01 ^b	86.86 ^c	0.84	<0.001
Ash	9-10	8.19 ^b	10.64 ^a	3-5	2.89 ^c	1.70 ^d	2.12	<0.001
CP	12-14	13.07 ^a	10.03 ^b	10-12	10.15 ^b	2.58 ^c	2.23	<0.001
CF	7-10	6.23 ^b	19.13 ^a	8-12	2.53 ^c	1.98 ^c	4.0	<0.001
EE	12-14	8.57 ^a	0.66 ^c	3-4	1.59 ^b	0 ^d	1.90	<0.001
Ca	0.06-0.10	0.31 ^a	0.31 ^a	0.02-0.05	0.21 ^b	0.31 ^a	0.02	<0.001
P	1.2-1.8	2.01 ^a	0.97 ^b	0.1-0.2	0.59 ^c	0.04 ^d	0.41	<0.001
Mg	0.6-0.7	0.96 ^a	0.46 ^b	0.2-0.3	0.19 ^c	0.10 ^d	0.19	<0.001
K	1.5-1.8	1.41 ^a	0.81 ^b	0.4-0.5	0.64 ^c	0.57 ^d	0.18	<0.001
GE	4,400-4,500	3,522.50	3,187.80	3,200-3,500	3,848.30	3,403.20		
(Kcal/kg)								
ME								
(MJ/kg)								
Poultry	2,900-3,100	2,994.13	2,550.24	1,200-1,400	2,613.22	3,199.01		
Swine	2,800-3,000	2,465.75	2,040.19	1,200-1,300	1,393.72			

⁻¹Nutrient Requirements of Swine and Poultry, National Research Council (NRC); DM-Dry Matter; CP-Crude Protein; CF-Crude Fiber; EE-Ether Extract or Crude Fat; Ca-Calcium; P-Phosphorous; Mg-Magnesium; K-Potassium; GE-Gross Energy; ME-Metabolizable Energy; Kcal/kg-Kilo calorie per kilogram; MJ/kg-megajoules per kilogram

-SEM-standard error of the mean

-Means with different superscript within a row are significantly different (p<0.005)

Leaf meal

Table 2. Comparison of the nutrient composition and energy contents of locally available leaf meals

Items (% DM)	Feedstuffs					SEM	P-value
	¹ Moringa	*Cassava leaves	Cassava leaves	Moringa	Madre de agua		
DM	89-91	85-88	89.40	89.12	88.58	0.24	0.207
Ash	**8-13	**7-10	6.8 ^c	12.92 ^b	20.65 ^a	4.01	<0.001
CP	25-30	20-25	24.18 ^b	27.11 ^a	16.51 ^c	3.16	<0.001
CF	7-10	10-15	14.16 ^a	8.97 ^c	12.69 ^b	1.54	<0.001
EE	**4-7	**2-4	4.19 ^a	4.21 ^a	1.79 ^b	0.80	<0.001
Ca	2-3	1-1.5	1.02 ^c	1.99 ^b	5.75 ^a	1.44	<0.001
P	0-0.5	0.2-0.3	0.30 ^c	0.37 ^b	2.02 ^a	0.56	<0.001
Mg	0.5-0.6	0.4-0.5	0.49 ^b	0.54 ^a	0.57 ^a	0.02	<0.001
K	1.5-2.0	1.0-1.5	1.41 ^b	1.44 ^b	2.06 ^a	0.21	<0.001
GE (Kcal/kg)	4,600-4,800	4,500-4,700	3,534.30	3,358.50	2,806.70		
ME (MJ/kg)							
Poultry							
Swine	1,500-1,800	1,200-1,400	2,994.13	3,523.04	2,550.24		
			2,465.75	3,187.52	2,040.19		

⁻¹Nutrient Requirements of Swine and Poultry, National Research Council (NRC);

⁻²Feedipedia

DM-Dry Matter; CP-Crude Protein; CF-Crude Fiber; EE-Ether Extract or Crude Fat; Ca-Calcium; P-Phosphorous; Mg-Magnesium; K-Potassium; GE-Gross Energy; ME-Metabolizable Energy; Kcal/kg-Kilo calorie per kilogram; MJ/kg-megajoules per kilogram

-SEM-standard error of the mean

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	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 8 of 8

Legume, Grains, and Sprouts

Table 3. Proximate analysis, mineral composition, and energy contents of locally available legumes, grains, and sprouts.

Items (% DM)	Feedstuffs											SEM	P-value
	¹ Soybean Meal	<u>Mungbean</u>	MBS (3d)	MBS (6d)	MBS (9d)	*Corn	Corn	CS (3d)	CS (6d)	CS (9d)	Soybean Whole		
DM	88-90	95.59 ^c	92.66 ^a	94.37 ^d	91.72 ^f	88-90	94.91 ^{cd}	97.77 ^b	97.92 ^b	98.79 ^a	90.87 ^a	0.94	<0.001
Ash	6-7	3.65 ^d	4.53 ^c	5.40 ^b	6.27 ^a	1.2-1.5	1.37 ^f	1.60 ^e	1.28 ^f	1.42 ^f	6.26 ^a	0.72	<0.001
CP	44-48	23.41 ^e	30.39 ^d	33.27 ^c	36.16 ^b	7-9	8.39 ^g	9.08 ^g	8.93 ^g	9.46 ^f	39.6 ^a	4.39	<0.001
CF	6-7	4.0 ^{ge}	7.47 ^c	8.47 ^b	13.65 ^a	2-3	3.53 ^{gh}	2.24 ^h	2.78 ^g	3.28 ^f	5.38 ^d	1.25	<0.001
EE	1.5-2	0.93 ^f	1.32 ^{ef}	1.30 ^{ef}	1.67 ^e	3-4	3.62 ^c	4.27 ^b	3.06 ^d	3.60 ^c	16.17 ^a	1.57	<0.001
Ca	0.20-0.30	0.11 ^d	0.29 ^c	0.44 ^{ab}	0.42 ^b	0.02-0.03	0.34 ^{bc}	0.29 ^c	0.29 ^c	0.34 ^{bc}	0.55 ^a	0.04	<0.001
P	0.60-0.70	0.29 ^d	0.48 ^c	0.54 ^b	0.64 ^a	0.25-0.30	0.24 ^{def}	0.25 ^{de}	0.19 ^f	0.23 ^{ef}	0.59 ^{ab}	0.05	<0.001
Mg	0.25-0.30	0.13 ^d	0.17 ^c	0.19 ^b	0.24 ^a	0.10-0.12	0.06 ^e	0.05 ^e	0.04 ^{ef}	0.04 ^{ef}	0.23 ^a	0.02	<0.001
K	2.00-2.50	1.17 ^d	1.37 ^c	1.63 ^b	1.77 ^a	0.35-0.40	0.42 ^e	0.41 ^e	0.32 ^e	0.36 ^e	1.65 ^b	0.20	<0.001
GE	4,400	3,721.6	3,590.7	3,632.3	3,507.4	4,300	3,922.6	4,069.1	4,003	4,078	4,194		

¹Nutrient Requirements of Swine and Poultry, National Research Council (NRC); DM-Dry Matter; CP-Crude Protein; CF-Crude Fiber; EE-Ether Extract or Crude Fat; Ca-Calcium; P-Phosphorous; Mg-Magnesium; K-Potassium; GE-Gross Energy; ME-Metabolizable Energy; Kcal/kg-Kilo calorie per kilogram; MJ/kg-megajoules per kilogram

-SEM-standard error of the mean

-Means with different superscript within a row are significantly different (p<0.005)

Marine By-products

Table 4. Comparison of the proximate, mineral, and energy contents of fish silage and shrimp meal

Items (% DM)	Feedstuffs				SEM	P-value
	*Shrimp Meal	Shrimp Meal	*Fish Meal	Fish Silage		
DM	90-92	92.30 ^a	90-92	86.08 ^b	3.11	<0.001
Ash	20-25	39.10 ^a	10-20	7.14 ^b	15.98	<0.001
CP	40-48	31.49	60-72	28.75	1.37	0.488
CF	5-7	20.19 ^a	1-2	14.23 ^b	2.98	<0.001
EE	6-8	0.61 ^b	8-12	2.27 ^a	0.83	<0.001
Ca	5-7	13.06 ^a	4-6	1.71 ^b	5.65	<0.001
P	2-3	1.90 ^a	2.8-3.5	1.12 ^b	0.39	<0.001
Mg	0.5-1.0	0.59 ^a	0.1-0.3	0.13 ^b	0.23	<0.001
K	0.5-1.0	0.40 ^b	0.3-0.5	0.64 ^a	0.12	<0.001
GE	4,500- 4,700	2,164.4	5,000- 5,200	3,271.1		

¹Nutrient Requirements of Swine and Poultry, National Research Council (NRC);

DM-Dry Matter; CP-Crude Protein; CF-Crude Fiber; EE-Ether Extract or Crude Fat;

Ca-Calcium; P-Phosphorous; Mg-Magnesium; K-Potassium; GE-Gross Energy;

ME-Metabolizable Energy; Kcal/kg-Kilo calorie per kilogram; MJ/kg-megajoules per kilogram

-SEM-standard error of the mean

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